

Date	30 September 2026	Team ID	SWTID-2026-26
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer	Maximum Marks	10 Marks

PROJECT DEVELOPMENT PHASE

Model Performance Test

Project team shall fill the following information in model performance testing template.

S.No.	Parameter	Values	Screenshot																				
1.	Metrics	Regression Model (Lead Time Estimation): - MAE: 2.14 hours - MSE: 6.82 hours² - RMSE: 2.61 hours - R² Score: 0.912 (91.2% explained variance) Classification Model (Queue & Task Routing): - Accuracy Score: 94.8% - Confusion Matrix: - Hardware TP: 712 (94.6%) Misallocated: 41 - ServiceDesk TP: 152 Software TP: 104 - Classification Report: - Hardware: Prec: 96.2%, Rec: 94.6%, F1: 0.954 - ServiceDesk: Prec: 84.4%, Rec: 87.9%, F1: 0.861 - Software: Prec: 88.9%, Rec: 91.2%, F1: 0.900 - Weighted Average: Prec: 94.6%, Rec: 94.8%, F1: 0.947	Model Evaluation Board • Metrics Summary Test Set: N=1,040 <table><tr><td>94.8% ACCURACY</td><td>0.947 MACRO F1</td><td>2.14h MAE</td><td>0.912 R² SCORE</td></tr></table> <table><tr><th>Actual \ Pred</th><th>Hardware</th><th>ServiceDesk</th><th>Software</th></tr><tr><th>Hardware (Laptop)</th><td>712 (94.6%)</td><td>28 (3.7%)</td><td>13 (1.7%)</td></tr><tr><th>ServiceDesk</th><td>14 (8.1%)</td><td>152 (87.9%)</td><td>7 (4.0%)</td></tr><tr><th>Software Triage</th><td>6 (5.3%)</td><td>4 (3.5%)</td><td>104 (91.2%)</td></tr></table>	94.8% ACCURACY	0.947 MACRO F1	2.14h MAE	0.912 R ² SCORE	Actual \ Pred	Hardware	ServiceDesk	Software	Hardware (Laptop)	712 (94.6%)	28 (3.7%)	13 (1.7%)	ServiceDesk	14 (8.1%)	152 (87.9%)	7 (4.0%)	Software Triage	6 (5.3%)	4 (3.5%)	104 (91.2%)
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2.	Tune the Model	Hyperparameter Tuning: - Technique: Bayesian Optimization & RandomizedSearchCV (100 iterations) - Pipeline: Gradient Boosted Tree Ensembles (XGBoost / LightGBM) - Optimized Parameters: - n_estimators: 250 (from [100, 200, 250, 500]) - max_depth: 6 (from [3, 4, 6, 8]) - learning_rate: 0.05 (from [0.01, 0.05, 0.1, 0.2]) - subsample: 0.85 colsample_bytree: 0.80 - confidence_threshold: 0.85 (cutoff to eliminate false queue dispatch) Validation Method: - Cross-Validation: 5-Fold Stratified Cross-Validation (K=5) - Data Split: 80% Training (4,160) / 20% Test (1,040) - Convergence: Early stopping patience at 15 rounds on validation loss.	Hyperparameter Optimization • 5-Fold CV Converged Graph showing accuracy vs iterations. Tuned CV: 94.8%, Baseline: 88.5%, Best Params (It 100). Fold Score Std Dev: ±0.006 Final Validation Accuracy: 94.8%																				

